



CRITICAL SYNCHRONIZATION LAWS IN PHYSICAL PERCEPTION:

- **Motion Blur Uncertainty:** Moving at velocity $v = 1.5 \text{ m/s}$ with $t_{\text{exp}} = 10 \text{ ms}$ blurs visual features across $\Delta x = v \cdot t_{\text{exp}} = 15 \text{ mm}$ of spatial smear.
- **Temporal Skew Illusion:** Correlating an image with an IMU reading using software timestamps creates an 11 ms phase mismatch, inducing catastrophic drift in Kalman state estimators.
- **PTP (IEEE 1588) Solution:** The image sensor triggers a hardware GPIO strobe at exposure start. The PTP timer registers the true physical midpoint ($t_{\text{mid}} = t_{\text{start}} + \frac{1}{2}t_{\text{exp}}$) with sub-microsecond precision.